Why the Moon Has Phases

A reading for grades 6–8 · VIABLE Lab example course

Look at the Moon on a few nights in a row and it seems to change shape. One night it is a thin curve, a week later it is half a circle, and about two weeks after the thin curve it is a full, bright disk. Then it shrinks again. These changing shapes are called the Moon's **phases**. The Moon itself never changes shape. What changes is how much of its lit side we can see from Earth.

The Moon makes no light of its own

The Moon is a ball of rock. It shines only because sunlight hits it and bounces off, the way a bicycle reflector shines when a headlight hits it. At every moment, the Sun lights exactly one half of the Moon: the half that faces the Sun. The other half is in darkness. This is true no matter where the Moon is in its path around Earth.

The Moon travels around Earth

The Moon orbits Earth about once every 29.5 days. As it moves, the angle between the Sun, Earth and the Moon changes. Sometimes we look at the lit half head-on, sometimes we see it from the side, and sometimes the lit half faces away from us almost completely.

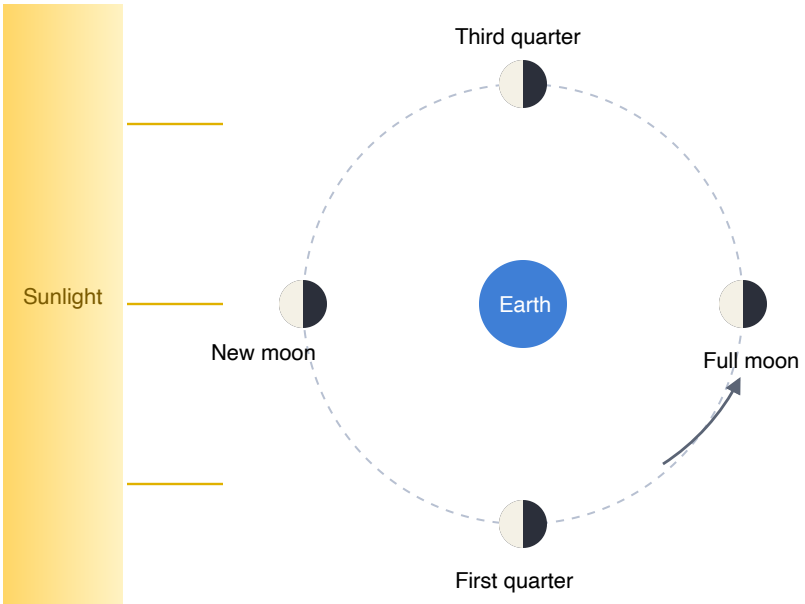


Figure 1. Seen from far above Earth's North Pole (not to scale), the Moon travels counterclockwise: new, first quarter, full, third quarter. The Sun always lights the half of the Moon that faces it; what we see from Earth depends on where the Moon is.

Following one cycle

1. **New moon.** The Moon is between Earth and the Sun. Its lit half faces away from us, so we see almost nothing.
2. **Waxing crescent.** A few days later, a thin sliver of the lit half comes into view. “Waxing” means growing.
3. **First quarter.** About a week after new moon, we see half of the Moon’s face lit. It is called a quarter because the Moon has gone a quarter of the way around its orbit.
4. **Waxing gibbous.** More than half of the face is lit, and the lit part keeps growing.
5. **Full moon.** Earth is between the Sun and the Moon, though slightly above or below the line between them. We see the whole lit half.
6. **Waning gibbous, third quarter, waning crescent.** The lit part we can see shrinks (“wanes”) through the same shapes in reverse, until the next new moon.

A common mistake: Earth’s shadow

Many people think the dark part of the Moon is Earth’s shadow. It isn’t. If Earth’s shadow caused the phases, a crescent would happen when Earth is between the Sun and the Moon, but that is when we see a *full* moon. Earth’s shadow does fall on the Moon now and then, during a **lunar eclipse**. Eclipses are rare because the Moon’s orbit is tilted a little, so the Moon usually passes just above or below Earth’s shadow.

Try it yourself

In a dark room, place a lamp on one side to be the Sun. Hold a ball at arm’s length to be the Moon; your head is Earth. Turn slowly in a circle, keeping the ball in front of you. Watch how much of the ball’s lit side you can see as you turn. You will see every phase, and the lamp never moves.

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